

scripts/check-canonical-root-and-upstreams.sh — User Guide

Last verified: 2026-05-15 (constitution-compliance plan Phase 8)

Inheritance: HelixConstitution §11.4.35 (Canonical-Root Inheritance Clarity) + §11.4.36 (Mandatory install_upstreams) + §11.4.18 (script docs)

Overview

Combined verifier for two related HelixConstitution clauses:

§11.4.35 — Canonical-Root Inheritance Clarity

Three sub-checks:

- **(a)** Consumer's root `CLAUDE.md` and `AGENTS.md` **MUST** open with an inheritance pointer (within the first 40 lines): either `## INHERITED FROM constitution/<file> heading` OR Claude Code's `@constitution/<file> import` syntax.
- **(b)** The constitution submodule's three canonical files **MUST** exist: `constitution/CLAUDE.md`, `constitution/AGENTS.md`, `constitution/Constitution.md`.
- **(c)** The constitution submodule's own `CLAUDE.md` and `AGENTS.md` **MUST NOT** carry `## INHERITED FROM` at top level — they **ARE** the canonical root and must not appear to inherit from elsewhere. **Pattern is allowed inside fenced `` code blocks** (where it's documentation showing consumers the inheritance pattern).

§11.4.36 — Mandatory install_upstreams script

Every owned-by-us submodule **MUST** ship an `install_upstreams` script at one of these locations:

- `<submodule>/install_upstreams.sh`
- `<submodule>/install_upstreams`
- `<submodule>/scripts/install_upstreams.sh`

- <submodule>/scripts/install_upstreams
- <submodule>/Upstreams/install.sh
- <submodule>/upstreams/install.sh

The script is what a freshly-cloned submodule uses to self-provision its own remote topology (per §6.W: github + gitlab minimum for basic-digital submodules; 4 mirrors for HelixDevelopment-owned submodules).

Per-submodule waivers are supported (INSTALL_UPSTREAMS_WAIVERS=() in the scanner) with mandatory rationale. No active waivers in 2026-05-15 baseline.

Modes

Flag	Behavior
--strict (default)	Exit 1 on any violation
--advisory	Exit 0 even on violation
LAVA_CANONICAL_ROOT_STRICT=0 env	Same as --advisory

Usage

```
bash scripts/check-canonical-root-and-upstreams.sh
# strict (default)
bash scripts/check-canonical-root-and-upstreams.sh --advisory
# exit 0 even on violation
LAVA_CANONICAL_ROOT_STRICT=0 bash scripts/check-canonical-root-and-upstreams.sh # env-driven
```

Hermetic test

tests/check-constitution/test_canonical_root_and_upstreams.sh — 6 fixtures:

1. test_compliant_fixture_passes — baseline compliant fixture
2. test_missing_root_pointer_rejected — root CLAUDE.md without inheritance pointer
3. test_canonical_self_inheritance_rejected — canonical root carrying its own ## INHERITED FROM outside any code block
4. test_missing_install_upstreams_rejected — submodule lacking install_upstreams
5. test_advisory_mode_returns_zero — advisory swallows exit code

6. `test_canonical_fenced_code_block_pointer_passes` — `## INHERITED`
FROM INSIDE a fenced ````` markdown block is documentation, not
violation (discrimination test for false-positive prevention)

Known violations (2026-05-15 audit)

§11.4.35 sub-checks all PASS in current Lava tree (after the Phase 8 false-positive fix in §11.4.35.c — the constitution submodule's documentation of the pointer pattern inside a fenced code block is correctly recognized as non-violating).

§11.4.36 — 10 of 16 owned submodules MISSING `install_upstreams`:

Submodule	Status
Auth	✓ has <code>install_upstreams</code>
Cache	✓ has <code>install_upstreams</code>
Challenges	✗ MISSING — owed in follow-up
Concurrency	✓ has <code>install_upstreams</code>
Config	✗ MISSING — owed in follow-up
Containers	✗ MISSING — owed in follow-up
Database	✓ has <code>install_upstreams</code>
Discovery	✗ MISSING — owed in follow-up
HTTP3	✗ MISSING — owed in follow-up
Mdns	✗ MISSING — owed in follow-up
Middleware	✗ MISSING — owed in follow-up
Observability	✓ has <code>install_upstreams</code>
RateLimiter	✗ MISSING — owed in follow-up
Recovery	✗ MISSING — owed in follow-up
Security	✓ has <code>install_upstreams</code>
Tracker-SDK	✗ MISSING — owed in follow-up

Resolution path: each submodule gains its own `install_upstreams.sh` in a per-submodule commit. Per-submodule cycle: write script → commit to submodule → push to GitHub + GitLab (per §6.W) → bump pin in parent Lava repo. The parent commit can batch the 10 pin bumps once each submodule has its own `install_upstreams`.

Cross-references

- HelixConstitution Constitution.md §11.4.35 + §11.4.36 (the mandates)
- docs/plans/2026-05-15-constitution-compliance.md Phase 8
- docs/helix-constitution-gates.md CM-CANONICAL-ROOT-CLARITY + CM-INSTALL-UPSTREAMS-RAN rows
- scripts/verify-all-constitution-rules.sh (wraps this scanner per §11.4.32)

2026-05-15 update — HelixQA waiver

Phase 4 of the constitution-compliance plan adopted submodules/helixqa (HelixDevelopment-owned QA orchestration framework) at upstream HEAD 403603db. HelixQA's CLAUDE.md / AGENTS.md / CONSTITUTION.md follow the canonical-root

INHERITED FROM Helix Constitution pointer pattern (HelixDevelopment-authored) rather than Lava's heading-anchored §6.R / §6.S / §6.X / §6.AD pointer-block format. HelixQA also lacks helix-deps.yaml + install_upstreams.sh wrapper script.

Resolution: HELIX_DEV_OWNED=("HelixQA") waiver list + is_helix_dev_owned() helper skip HelixQA in every per-Submodule loop in this scanner. Waiver entries cite Phase 4-debt: PR to HelixDevelopment/HelixQA upstream owed to add the missing files. Once upstream merges + Lava's pin advances to include them, HelixQA can be removed from the waiver list.

2026-05-16 update — HelixQA waiver RESOLVED

The Phase 4-debt PR to HelixDevelopment/HelixQA upstream landed as commit b13ba7c (feat(gov): add helix-deps.yaml + install_upstreams.sh wrapper), adding both missing files at HelixQA's repo root. Lava's pin advanced to b13ba7c in the same parent commit that removed HelixQA from INSTALL_UPSTREAMS_WAIVERS. The waiver list is now empty — the per-Submodule loop in this scanner treats HelixQA on equal terms with the other 16 owned submodules; all 17 must ship a recognizable install_upstreams.sh wrapper invoking Upstreams/*.sh (or upstreams/*.sh) recipes. The INSTALL_UPSTREAMS_WAIVERS=() empty array is intentionally retained as a forensic anchor. The §11.4.35 canonical-root inheritance pointer audit for HelixQA's CLAUDE.md/

AGENTS.md/CONSTITUTION.md remains unaffected by this closure — HelixQA's HelixDevelopment-authored `## INHERITED FROM Helix Constitution` pointer pattern continues to satisfy the constitution-submodule's own inheritance discipline, and the §11.4.35 (a) check passes for HelixQA without needing waivers (its CLAUDE.md opens with the inheritance pointer, just sourced from HelixDevelopment's pattern rather than Lava's §6.AD block — both are valid).

2026-09-19 update — PERMANENT superspec waiver (third-party, not a debt item)

submodules/superspec (upstream github.com/WangX0111/superspec, a spec-kit extension consolidated onto the project's standard submodules/path this same day per an operator decision — see `scripts/check-constitution.sh`'s companion doc for the sibling `THIRD_PARTY_OWNED` fix, commit 80b1bb2c) tripped §11.4.36 the moment it entered the `submodules/*/scan`: it is not a vasic-digital or HelixDevelopment repo, so there is no upstream Lava can PR an `install_upstreams.sh` wrapper into — unlike the HelixQA case above, this is not a debt item awaiting an upstream merge; it is a structural non-applicability. Added `"superspec:..."` to `INSTALL_UPSTREAMS_WAIVERS` with a rationale citing this. Falsifiability rehearsed: removing the waiver entry reproduces §11.4.36 missing `install_upstreams`: `superspec` verbatim; restoring it passes again. This waiver is expected to remain indefinitely — there is no resolution path to track, unlike the now-empty HelixQA entry above.